

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing

Online Pharmacy Store -Final Document

Module: SE1020 – Object Oriented Programing| **Group ID:** [Group 10]

Group Members

	Full Name	Student Id	Assign Function
1	Soysa C.N	IT21821868@my.sliit.lk	Pharmacy Management
2	Dilshani H.T.D.P	IT21379574@my.sliit.lk	Payment Management
3	Diasanayake .J .S	IT21318870@my.sliit.lk	Admin
4	Gamage H.G.U.E	IT21831522@my.sliit.lk	Customer

Component 01 – Pharmacist

Description

The Pharmacist manages the medicine store. They can add, update, or remove medicines and ensure correct stock availability.

CRUD Operations

Create - Add new medicines to store

Read - View all medicines

Check stock levels

Update - Edit medicine details (price, description, quantity)

Delete - Remove medicines from store

UI Components

- Dashboard
- Add Medicine Form
- Medicine List Table
- Edit Medicine Page

OOP Concepts Applied

- Encapsulation- Medicine details (name, price, stock) are stored in objects.
- Inheritance- Pharmacists inherit from User.
- Polymorphism- Manage Inventory behaves differently from other roles.
- Abstraction- Pharmacists interact with UI without database knowledge.

Component 02 – Accountant

Description

The accountant is responsible for managing all financial activities in the system. They monitor payments made by customers, verify transactions, generate financial reports, and handle payment issues such as cancellations or refunds. This role ensures the financial accuracy and integrity of the online pharmacy system.

CRUD Operations

Create Generate financial reports

Record transaction adjustments (if needed)

Read

View all payment records

View transaction history

View customer payment details

Update

Update payment status (e.g., verified, refunded)

Modify transaction records if errors occur

Delete

Cancel payments

Remove invalid or duplicate transaction records

UI Components

- Receipt ledger

OOP Concepts Applied

- Inheritance- Accountant inherits from the base User class.
- Polymorphism - Different payment methods (Card, Cash, Online).
- Abstraction- Complex financial processing is hidden behind simple UI operations.

Component 03 – Admin

Description

The admin manages the entire system, including users, pharmacists, and transactions. They have full control over system operations.

CRUD Operations

Create - Add new users or pharmacists

Read -View all users

View system reports

View payment records

Update - Modify user roles/details

Delete -Remove users

Cancel transactions

UI Components

- Admin Dashboard
- User Management Panel
- Medical panel
- Payment Monitoring Panel

OOP Concepts Applied

- Encapsulation-Admin data and permissions are secured.
- Inheritance-Admin inherits from User.
- Polymorphism-Admin functions override base user behaviors.
- Abstraction-Complex system management simplified through dashboard.

Component 04 - Customer

Description

The Customer is the main user of the system who can browse medicines, add items to the cart, and complete purchases through the payment portal. Customers interact with the system to place orders and manage their shopping experience.

CRUD Operations

Create -Registering a new account

Add medicines to cart

Place an order

Read -View medicine list

View cart items

View order history

Update -Update profile details

Modify cart (change quantity)

Delete -Remove items from cart

Cancel orders

UI Components

- Registration/Login Form
- Medicine Store Page
- Cart Page
- Checkout Page
- Order History Page
- Profile Settings Page

OOP Concepts Applied

- Encapsulation-Customer data (name, email, cart) is stored securely inside the class.
- Inheritance-Customer inherits from a base User class.
- Polymorphism-Methods like view Details behave differently for Customer vs Admin.
- Abstraction-Customer interacts with simple UI without seeing backend logic.

Class Diagram

